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# OP0262 INTEGRATED MULTI-OMICS ANALYSIS REVEALING SUBTYPE-SPECIFIC PATHOGENIC MECHANISMS IN IDIOPATHIC INFLAMMATORY MYOPATHIES

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## Abstract

### Background:

Idiopathic inflammatory myopathies (IIM) are a diverse group of systemic autoimmune disorders, with muscle weakness as the primary clinical feature. Transcriptomic analysis of muscle tissues have revealed that each IIM subtype has a distinct gene expression profile. However, the key pathways involved in these subtypes remain largely unexplored, and their relationships with clinical indices have yet to be established.

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In this study, we analyzed multi-omics data, including genomic (exonic and intronic), proteomic, and metabolomic profiles of IIM muscle tissues, to investigate their roles in the pathogenesis of IIM.

## Methods:

RNA sequencing, liquid chromatography-tandem mass spectrometry and untargeted metabolomics were conducted on muscle tissues obtained from 159 IIM patients (91 DM, 27 IMNM, 41 ASS) and 18 controls. The transcript reads at the exon and intron levels were calculated using FeatureCounts and iREAD, respectively. A differential expression analysis was conducted using R package edgeR, DEP and MetaboAnalystR. The differentially expressed exon (DEE), intron (DEI), protein (DEP) and metabolite (DEM) were integrated using Multi-Omics Factor Analysis (MOFA2). Pathway enrichment was performed using clusterProfiler. The correlation between exon, intron, protein, metabolite and clinical features was evaluated using Spearman, Pearson or point-biserial correlation tests. Kaplan-Meier analysis was used to compare the time to recurrence in patients with different expression levels of exons/introns/proteins/metabolites.

## Results:

Single-omics data alone are insufficient for differentiating between IIM subtypes. However, combining multimodal data through MOFA2 enables a more comprehensive representation of the characteristic features of these subtypes. We identified 344 upregulated and 454 downregulated features specifically associated with dermatomyositis (DM). For immune-mediated necrotizing myopathy (IMNM), we found 748 subtype-specific features, including 597 upregulated and 151 downregulated features. In antisynthetase syndrome (ASS), 297 features were identified, comprising 151 upregulated and 146 downregulated features. The most prominent pathways in DM included interferon (IFN)-related pathways, the cytokine-mediated signaling pathway, and the IKK/NF-κB signaling pathway. In IMNM, extracellular matrix-related pathways, along with pathways involved in macrophage migration and NK-T cell differentiation, were enriched. In ASS, metabolism-related and vascular-related pathways were predominant. The IFN and cytokine pathways were positively correlated with skin manifestations such as heliotrope rash, Gottron's papules, V signs, shawl signs, and holster signs, as well as with the Manual Muscle Testing (MMT8) score. Conversely, these pathways were negatively correlated with biochemical markers of muscle injury, including ALT, AST, LDH, CK, and MB. Furthermore, the IFN and cytokine pathways were negatively associated with muscle fiber necrosis and regenerating fibers. In contrast, pathways related to actin filament assembly (specifically MYH9), extracellular matrix organization, ketone metabolism, and nucleoside metabolism were positively correlated with muscle fiber necrosis and regenerating fibers, as well as with the same

ISG15_protein, MYL2_protein, ANXA1_protein, and D-Serine were effective in distinguishing patients based on their time to recurrence. The protein expression level of ANXA1 had the highest hazard ratio, followed by the exon expression of MX1.

## Conclusion:

Our study provides a comprehensive understanding of IIM pathogenesis through extensive multi-omics data analysis. We identified key genes and metabolites associated with IIM subtypes, characterized distinct biological pathways linked to clinical features, and proposed several potential biomarkers.

## REFERENCES:

**NIL**.

## Acknowledgements:

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## Disclosure of Interests:

**None declared**.

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| **Speaker information** | **Prof. Dr. Honglin Zhu** (Changsha, China) |
| **Room** | **B1** |
| **Date** | Friday, 13 June, 11:06 - 11:15 CEST |
| **Part of session** | > Basic Abstract Sessions: New ways of patient stratification with Systemic Diseases |

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